

User Guide and System Walkthrough

Stepwise operating instructions, internal processing details, and native vector architecture diagrams for the Streamlit dashboard, analysis workflow, MCP server, and optional LLM explanation layer.

IMPORTANT: An anomaly is unusual under a stated model. It is not proof of fraud, misconduct, or an incorrect election outcome. Aggregate precinct analysis is not a risk-limiting audit of ballot evidence.

Part I - Stepwise user instructions

1. Install

Open PowerShell in the repository. Create and activate a supported Python 3.11-3.13 environment, then install the dashboard, MCP, and development groups.

```
cd E:\election-analysis
py -3.12 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -e ".[dashboard,mcp,dev]"
```

For OpenAI summaries, also install the optional LLM group:

```
python -m pip install -e ".[llm]"
```

2. Prepare the CSV and mapping

Use official precinct-level results and keep the original file separately. Edit data.schema in config.yaml so each election concept points to its source column.

Concept	What it controls
jurisdiction	County, municipality, or reporting area
precinct	Stable precinct or voting-location identifier
valid_contest_votes	Candidate-share denominator
candidates	Source column, display label, and stable key
registered_voters + ballots_cast	Calculated turnout
vote_type	Mail, Early, Election Day, Provisional isolation
down_ballot_pairs	Same-party presidential/down-ballot comparison
latitude + longitude	Coordinate-based spatial analysis and marker maps

Never invent missing counts or coordinates. Unknown source columns are preserved.

Operate the dashboard

3. Start the dashboard

Run the Streamlit entry point, then open localhost:8501.

```
streamlit run src/dashboard.py
```

4. Load and validate

Load the fictional sample or upload an official CSV. Read validation findings, accepted count, excluded count, and every exclusion reason before analysis.

- Correct source data or the mapping outside the result table, then re-upload.
- Do not interpret an excluded record as a normal observation.

5. Choose one scope

Select candidate, methods, jurisdictions, vote types when available, turnout range, and minimum ballots. The displayed scope count is the exact analysis/export scope.

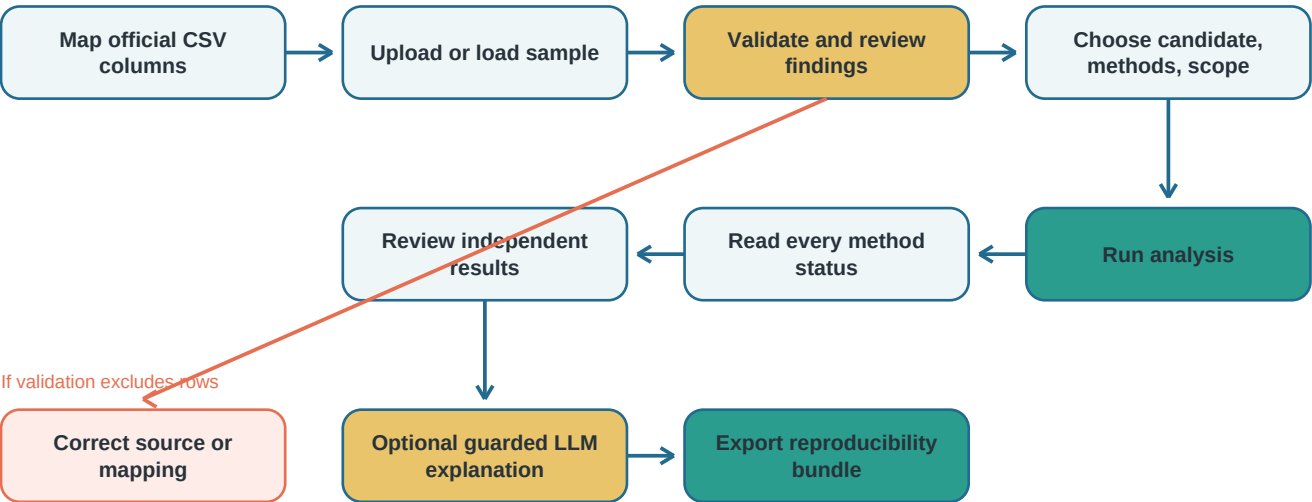
6. Run and read status first

Click Run selected analysis. Successful means output exists; unavailable means preconditions were not met; skipped means disabled or not selected; failed means a surfaced error occurred. There is no composite fraud score.

7. Review each result independently

Read its assumptions and limitation before comparing it with another method. A shared flag can motivate source verification, but does not establish a cause.

Operator journey



Interpret the ETA-compatible views

1. Down-ballot difference

The formula is $100 * (\text{presidential votes} - \text{same-party down-ballot votes}) / \text{presidential votes}$. Negative values mean the down-ballot candidate received more. Roll-off, split-ticket voting, eligibility, candidate effects, and vote type matter.

2. Vote share by vote count

Candidate share appears vertically and that candidate's precinct vote count appears horizontally. The fitted line is descriptive. The horizontal axis is not time.

3. Turnout analysis

The dashboard shows share versus turnout and aggregates candidate vote totals into turnout ranges. A tail or non-bell-shaped mixture is not independently proof of error.

Additional diagnostics

- Turnout/share residuals: leave-one-out deviations from an exploratory baseline.
- Digit tests: dataset-level tests with multiple-testing correction.
- Spatial tests: permutation Moran statistics; normal geography can cluster.
- Isolation Forest: unusual feature combinations, not fraud probabilities.
- DBSCAN: disabled by default; requires jurisdiction-specific calibration.

Coordinate-only maps are labeled marker maps. Polygon adjacency and choropleths require real polygon geometry and are never silently approximated.

Use LLM assistance, exports, and MCP

8. Enable optional LLM explanation

Set OPENAI_API_KEY, install the llm extra, and set llm.enabled to true. The model receives computed statuses, diagnostics, aggregate flag counts, and guardrails - not authority to create statistics or infer a cause.

- Allowed: explain assumptions, compare computed methods, ask contextual questions.
- Prohibited: invent significance, infer fraud, confirm an outcome, prioritize an audit.

9. Export a complete run

Download the reproducibility ZIP and keep it intact. It contains accepted input, scoped results, exclusions, flags, diagnostics, validation, configuration, source hash, random seed, and a factual report.

10. Start the custom MCP server

Use the installed command or module entry point. The default transport is stdio.

```
election-analysis-mcp
# equivalent
python -m src.mcp_server
```

- health - version, method list, and interpretation boundary
- sample_csv - bounded fictional CSV generation
- validate_csv - provenance, issues, exclusions, and bounded preview
- analyze_csv - deterministic methods and bounded result records
- narrative_context - computed facts and explicit LLM prohibitions

11. Verify the installation

Run lint, static typing, and the line/branch coverage suite.

```
ruff check src test
mypy src
pytest --cov=src --cov-branch --cov-report=term-missing
```

Part II - How the system works

1. Configuration becomes a contest schema

Defaults and config.yaml are safely merged. Candidate keys, down-ballot references, model settings, and spatial choices are validated before data processing.

2. Ingestion preserves source evidence

Bounded CSV bytes are decoded, hashed with SHA-256, copied without dropping unknown columns, mapped to canonical fields, and assigned stable IDs. Vote type joins the ID when mapped.

3. Validation separates accepted and excluded records

Missing or duplicate identifiers, invalid counts, impossible relationships, bad coordinates, and turnout discrepancies are reported. Critical errors exclude the row with reasons. No count or coordinate is imputed.

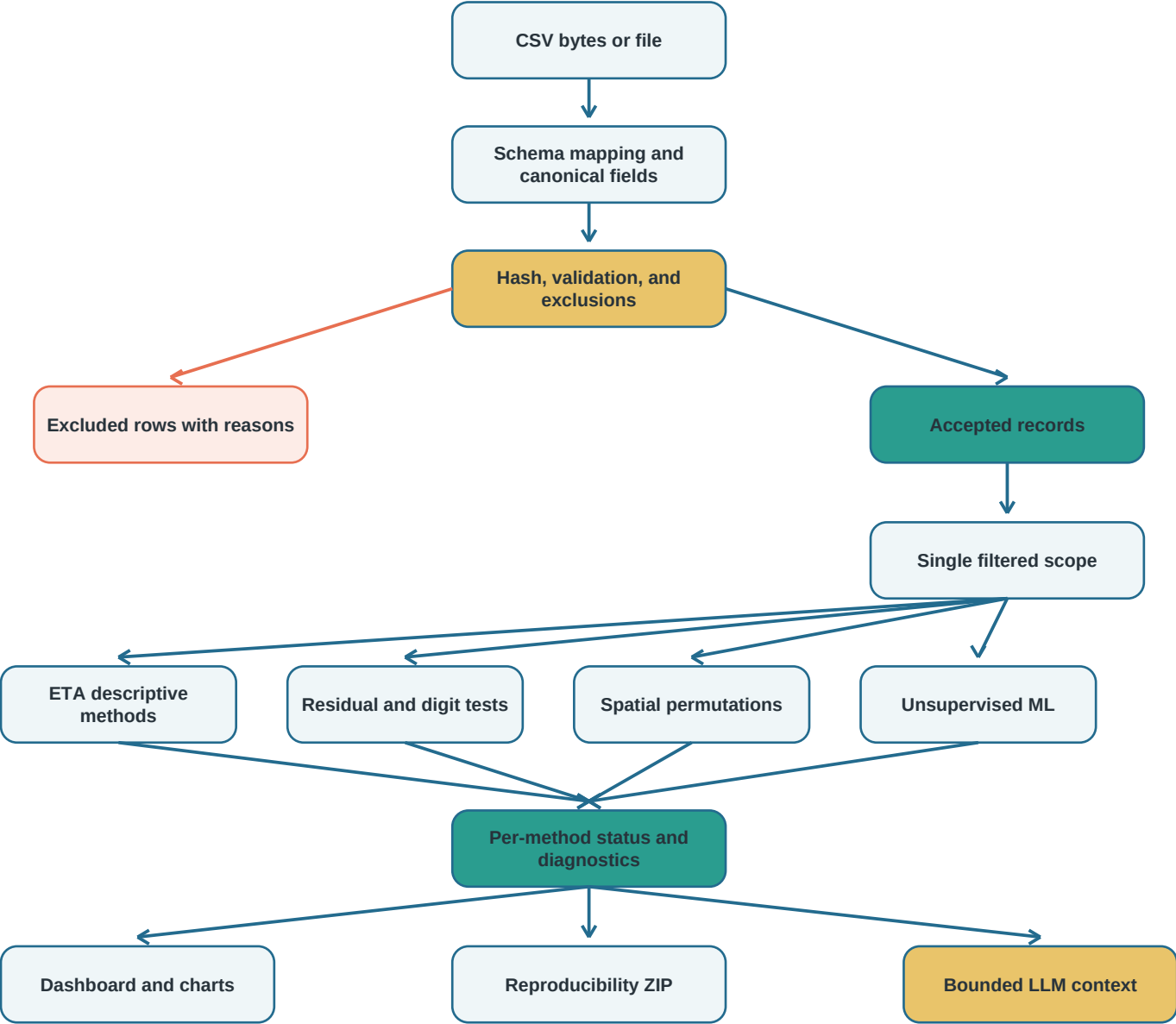
4. Filtering creates one scope

Jurisdiction, vote type, turnout, and ballot filters produce a copied frame. The same frame drives analysis, display, and export. A settings signature clears stale results.

5. Methods run in isolation

Every requested method returns a visible status. Unavailable or failed methods never become zero scores, and independent results are not collapsed into a composite.

Validated analysis pipeline



Method internals

6. Statistical outputs retain assumptions

Descriptive methods do not create automatic flags. Residual output stores baseline, prediction interval, leverage, and threshold. Digit tests store sample size, statistic, raw and adjusted p-values. Spatial output stores weights, permutations, and FDR control.

7. ML is deterministic and leakage-aware

An allow-list excludes prior result labels and free text. Training medians and robust scaling are reused. Fixed seeds stabilize Isolation Forest. DBSCAN rejects degenerate all-noise or zero-cluster results.

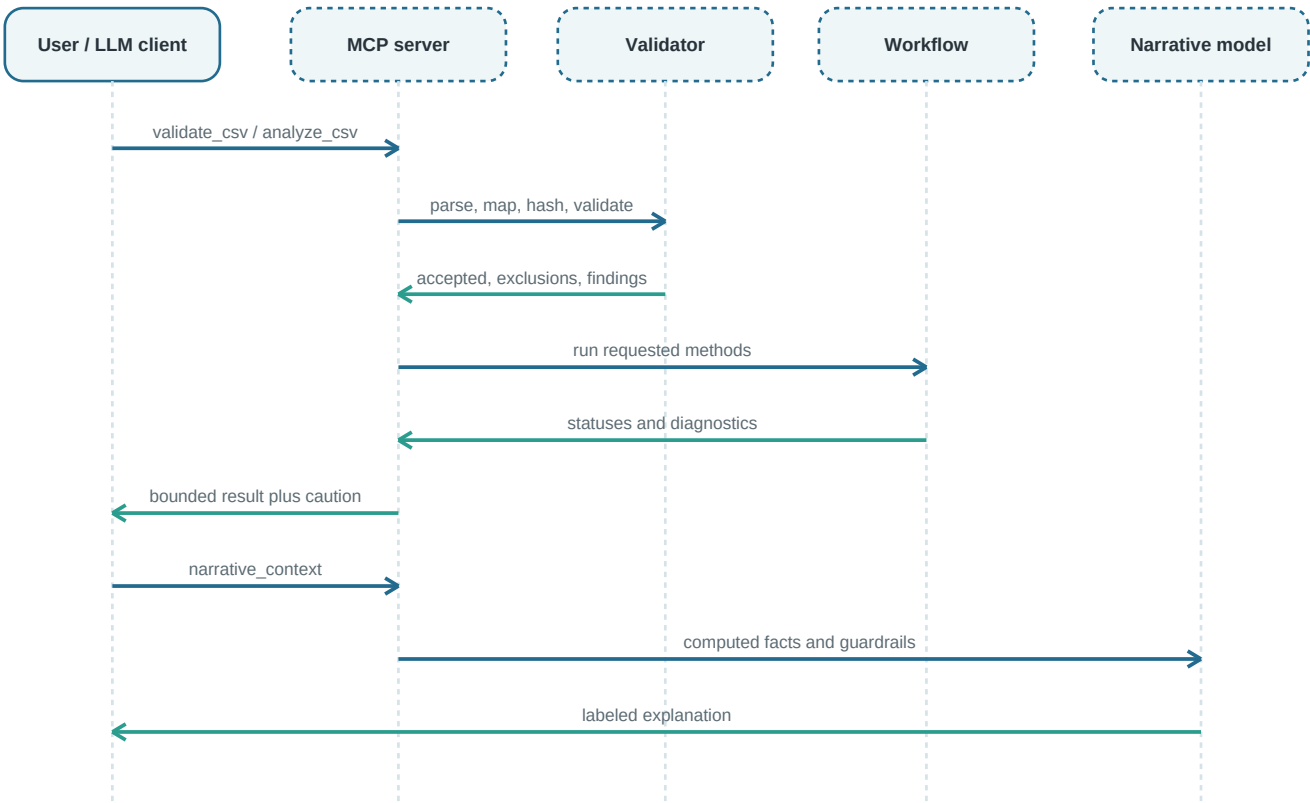
8. Visualization reflects available geometry

Plotly renders turnout, residual, vote-count, down-ballot, distribution, and marker views. Queen/rook adjacency and choropleths require polygons. KNN remains labeled KNN.

9. The LLM stays downstream

Numeric analysis completes without a model. Narrative input contains computed facts and guardrails. A provider failure returns a status and cannot alter results.

MCP and LLM sequence



Reproducibility and troubleshooting

10. Export closes the loop

The ZIP shows what was supplied, accepted, excluded, run, unavailable, and configured. Preserve it as the review artifact rather than copying only a chart or flag list.

Common problems

- No records remain: inspect exclusions, correct the source or mapping, and re-upload.
- Turnout unavailable: map registration and ballots, or explicit reported turnout.
- Down-ballot unavailable: configure the selected candidate's pair and count column.
- Spatial unavailable: provide coordinates, or real polygons for queen/rook weights.
- AI unavailable: install the llm extra, set OPENAI_API_KEY, and enable it.
- Settings changed: rerun; stale output is intentionally cleared.

Interpretation checklist

- Verify the source hash and validation report.
- Confirm analysis scope, vote type, candidate, and method status.
- Read the method limitation before discussing a pattern.
- Seek administrative, geographic, demographic, and historical context.
- Use trustworthy ballot evidence and an authorized audit design for outcome claims.

This system is an exploratory review tool. Its strongest contribution is transparent, reproducible triage and explanation - not a statistical shortcut to a fraud claim.

Further reading

See docs/methodology.md for scientific qualifications and literature references.